COMBINATORICS
NOTES JANUARY 29, 2026

1. Sterling Numbers. Con't

Let $f : M \to B$ be a function where M is a set of marbles and B is a set of boxes. Also let $m = |M|$ and $b = |B|$ We concern ourselves with the following counts.

(1) We want the count of the number of relations from M to B:

$$\text{count} = 2^{|M \times B|} = 2^{|M||B|} = 2^{mb}$$

(2) We want the count of the number of general functions from M to B which follows as:

$$\text{count} = b^m$$

(3) With $m \le b$ we want to count the number of one-to-one functions from M to B which follows as:

$$\text{count} = {}_bP_m$$

(4) With $m \le b$ we want to count the number of onto functions from M to B, which follows as:

$$\begin{aligned} \text{count} &= b^m - \binom{b}{1}(b-1)^m + \binom{b}{2}(b-2)^m - \binom{b}{3}(b-3)^m + \cdots + (-1)^b\binom{b}{b}(b-b)^m \\ &= b! \cdot \frac{1}{b!}\sum_{k=0}^{b}(-1)^k\binom{b}{k}(b-k)^m = b!\mathcal{S}(m,b) \end{aligned}$$

(5) With $m = b$ we want to count the number of bijections from M to B. This follows as

$$\text{count} = m! = b!$$

Now we consider the diagram

Date: January 30, 2026.

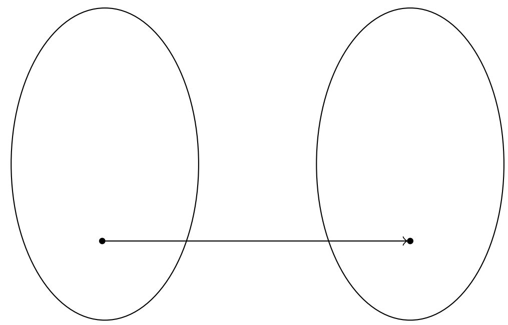

FIGURE 1. Set mapping

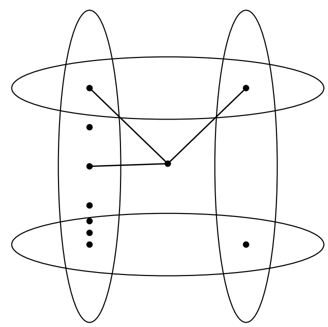

FIGURE 2. Something here